

AKSHAT ARYA

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[linkedin](#) — [leetcode](#) — [github](#) — [codeforces](#)

PROFESSIONAL SUMMARY

Final-year Electronics and Telecommunication Engineering student with a strong foundation in systems programming, computer networks, operating systems, and data structures. Experienced in developing distributed systems, storage systems, browser infrastructure, and developer tools using C++, Go, Python, and JavaScript.

EDUCATION

Jadavpur University

B.E in Electronics and Tele-Communication Engineering

Kolkata, West Bengal

Nov 2023 – May 2027

St. Tulsi Global School

CBSE Board Examinations

Azamgarh, Uttar Pradesh

2022

St. John's School

ICSE Board Examinations

Jaunpur, Uttar Pradesh

2020

EXPERIENCE

Software Developer Intern

Mar 2025 – Aug 2025

Aarish Technologies, Remote — C++, Pthreads, Ffmpeg, Filesystems, Python, CNNs

- Built a multi-IP camera system using CNNs for live video analytics and real-time database storage
- Designed a scalable C++ server capable of handling 200+ live camera feeds with 30ms latency on A6000 GPUs
- Achieved 80x speedup and 48% more efficient GPU usage through pipelining at similar hardware dependencies
- Integrated alert and monitoring modules enabling instant alerts with 2s response time in edge environments

MLRG Research Intern

June 2025 – July 2025

Indian Statistical Institute, Kolkata — VAEs, SMOTE, Classifiers — Dr. Swagatam Das

- Worked on Variational Autoencoders to improve SMOTE-style oversampling with imbalance ratios up to 1:100
- Created a binary based input system for VAEs, trading higher training cost for improved inference and latent space.
- Improved F1-score by 14.23%, end-to-end processing speedup of 4.74×, and latent clustering by 41%.

PROJECTS

Tab Recommender System [github](#) — [demo](#)

December – Jan 2026

Chrome Extension (Manifest V3) — JavaScript, Python, Chrome Extension APIs, Markovian Chains

- Developed a Chrome Extension that predicts the next websites and enables a 2-step shortcut to transfer instantaneously
- Implemented a 6-dimensional Markov model that predicts next 4 most probable browser tabs with constant-time inference.
- Built a daily offline training pipeline with efficient pre-processing and state reduction to prevent Markov state explosion.

Whack-A-Storage [github](#) — [demo](#)

July 2026

Distributed File Storage System — Go, TCP, AES-CTR, Content-Addressable Storage

- Built a distributed storage system in Go using custom TCP-based peer-to-peer networking for decentralized file storage.
- Leveraged goroutines and channels to concurrently manage peer communication, file transfers, and storage operations.
- Implemented CAS using hashed directory trees and AES-CTR encryption for deterministic lookup and secure replication.

Worse-Than-Git(WTGit) [github](#) — [demo](#)

Feb 2025 – Mar 2025

Version Control System — C++17, OpenSSL, STL, Filesystem

- Built a version control system supporting init, add, commit, log, checkout, revert, and repository status tracking.
- Implemented SHA-1 object storage, commit snapshots, and repository metadata using OpenSSL and C++17 Filesystem.
- Developed a class-based VCS with a fully functional CLI, robust validation, and fail-safe operations.

vNES [github](#) — [demo](#)

Mar 2026 – Apr 2026

Virtual NES Emulator — C++17, PixelGameEngine, OpenGL, ALSA, X11

- Implemented a cross-platform, fully playable NES emulator in C++, with 6502 CPU, PPU, APU and cartridge loading.
- Worked on a modular hardware abstraction with support for multiple cartridge mappers, real-time rendering, audio playback, and graphics debugging tools.
- Developed a custom ROM loader with multi-game support, extended mapper compatibility, and save/load state.

TECHNICAL SKILLS

- **Programming Languages:** C/C++, Python, Go, Verilog, JavaScript;
- **Frameworks/Tools:** Git, Linux, React, Node.js, Flask, Chrome Extension APIs, OpenSSL,
- **Relevant Coursework:** Data Structures and Algorithms, Operating Systems, Computer Organization and Architecture, Digital Logic and Circuits, Linear Algebra
- **Spoken Languages:** English, Hindi, Bengali

ACHIEVEMENTS

• **WBJEE 2023** – AIR 322

COMEDK 2023 – AIR 1018

JEE Main 2023 – 98 Percentile